

SUKUMO, Japan

WMO: 478970

Lat: 32.9206N

Lon: 132.6950E

Elev: 11

StdP: 101.19

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.3	1.3	-7.4	2.0	5.9	-6.1	2.2	6.0	9.5	7.4	8.5	7.7	2.5	60	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.0	32.0	25.3	31.0	25.4	30.2	25.2	26.7	29.8	26.2	29.3	25.8	29.0	4.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.1	28.6	25.3	20.5	28.3	24.9	19.9	28.1	83.7	29.7	81.7	29.5	79.9	29.1	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.3	7.0	6.1	-1.4	35.1	0.8	1.2	-2.0	36.0	-2.5	36.7	-3.0	37.4	-3.6	38.3
			-2.8	27.1	0.9	0.7	-3.4	27.5	-3.9	27.9	-4.4	28.3	-5.0	28.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	7.3	8.2	11.1	15.5	19.6	22.7	26.7	27.6	25.0	20.0	14.7	9.6
	DBStd	7.42	2.61	3.16	3.00	2.78	1.90	2.00	1.64	1.31	2.11	2.64	3.14	2.98
	HDD10.0	228	91	68	22	1	0	0	0	0	0	0	2	44
	HDD18.3	1346	342	282	223	91	9	0	0	0	0	13	116	271
	CDD10.0	2925	7	18	57	166	299	380	518	547	449	311	141	32
	CDD18.3	1002	0	0	0	5	49	130	259	288	199	66	5	0
	CDH23.3	8056	0	0	1	6	127	591	2478	3094	1508	246	4	0
	CDH26.7	2198	0	0	0	0	7	59	705	1063	346	18	0	0
Wind	WSAvg	2.8	2.9	2.9	3.0	3.1	2.9	2.7	2.6	2.8	2.7	2.6	2.8	3.0
Precipitation	PrecAvg	2045	74	93	144	164	198	300	228	235	270	163	109	67
	PrecMax	2717	181	238	269	311	402	528	676	640	597	425	298	130
	PrecMin	1394	23	35	78	76	83	112	48	20	68	29	26	15
	PrecStd	382	43	49	48	62	77	115	143	159	152	113	62	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.7	19.7	21.1	24.3	27.5	29.4	34.1	34.6	31.7	28.4	23.9	19.7
		MCWB	14.0	16.0	15.2	17.2	18.0	23.7	24.8	24.7	24.4	23.2	19.3	15.5
	2%	DB	15.1	17.8	19.6	22.8	25.7	28.2	31.6	32.5	30.4	26.9	22.6	17.9
		MCWB	11.0	14.3	15.1	17.1	18.3	23.5	25.5	25.4	24.4	22.1	17.8	13.6
	5%	DB	13.7	15.8	18.4	21.8	24.7	27.2	30.6	31.4	29.5	25.7	21.4	16.2
		MCWB	9.4	11.8	13.9	16.5	18.7	23.0	25.4	25.5	24.1	20.5	16.9	11.9
	10%	DB	12.3	14.0	17.0	20.7	23.8	26.3	29.8	30.6	28.7	24.7	20.1	14.9
		MCWB	8.4	9.9	12.5	15.8	18.6	22.4	25.2	25.4	23.7	19.7	15.9	10.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.2	17.1	18.1	20.7	22.6	25.5	27.3	27.4	26.6	24.9	21.2	17.6
		MCDB	17.0	19.0	19.9	22.4	24.4	27.8	30.5	30.5	28.8	27.1	22.9	18.8
	2%	WB	12.4	15.0	16.2	19.3	21.7	24.8	26.5	26.8	25.9	23.4	19.2	14.4
		MCDB	14.1	17.2	18.3	21.3	23.4	27.1	29.7	30.0	28.6	25.8	21.2	16.5
	5%	WB	10.4	12.8	14.9	18.0	20.8	24.2	26.0	26.4	25.3	22.2	17.9	12.9
		MCDB	12.7	15.2	17.3	20.5	22.8	26.5	29.2	29.6	28.1	24.5	20.4	15.4
	10%	WB	8.8	10.6	13.5	16.9	19.8	23.4	25.6	26.1	24.7	20.9	16.8	11.4
		MCDB	11.8	13.3	16.5	19.7	22.4	25.5	28.7	29.3	27.6	23.6	19.3	14.2
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.2	8.6	8.4	7.8	5.7	5.5	6.0	6.6	7.8	8.1	7.9
		MCDBR	9.8	9.7	9.9	9.6	9.3	6.8	6.9	7.4	7.1	8.0	8.5	9.5
		MCWBR	7.0	7.3	7.0	5.9	5.0	3.3	2.8	2.8	3.2	4.3	5.5	6.3
	5% WB	MCDBR	8.3	8.5	8.0	7.1	6.1	4.7	5.7	6.0	5.7	6.1	6.9	7.8
		MCWBR	7.3	7.5	7.2	5.6	4.8	2.9	2.7	2.7	3.0	4.0	5.3	6.2
Clear-Sky Solar Irradiance	taub	0.405	0.448	0.507	0.532	0.541	0.554	0.545	0.522	0.492	0.459	0.420	0.395	
	taud	2.170	2.073	1.925	1.917	1.964	2.021	2.086	2.159	2.199	2.194	2.224	2.211	
	Ebn at Noon	785	788	772	773	769	756	761	773	779	771	765	771	
	Edh at Noon	120	146	183	192	185	174	163	149	138	128	113	110	
All-Sky Solar Radiation	RadAvg	2.51	3.14	4.17	4.91	5.18	4.32	5.35	5.43	4.35	3.47	2.65	2.25	
	RadStd	0.22	0.26	0.30	0.49	0.55	0.61	0.64	0.46	0.43	0.36	0.26	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only		N/A	N/A	+0.86	N/A	+0.30	N/A	N/A	+76	N/A	N/A
Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page